Predicting Cancer Diagnosis Using Support Vector Machines

National Health Interview Survey(NHIS) 2022 | EDWIN OKWOR | SEATTLE UNIVERSITY

According to the World Health Organization, Cancer is the leading cause of death worldwide, accounting for about 10 million deaths in 2020. Cancer is a disease in which some of the body's cells grow uncontrollably and spread to other parts of the body. As per Standford Medicine health care, there are no single cause for cancer. It's often an interaction of many factors together that cause cancer. Some of the risk factors that could contribute include: Lifestyle, family history/inheritance, exposure to certain viruses, and environmental exposures. The main objective of this project is to predict whether a person will have cancer or not based on some demographic, body metric, life habit features.

01 Introduction

Can everyday health behaviors predict cancer? Using NHIS 2022 data, this project applies SVM models with three kernel types — linear, radial, and polynomial — to classify whether a respondent has ever been diagnosed with cancer based on demographics and lifestyle habits.

02 Dataset & Variables

Target: Cancer_Status (ever diagnosed)
Predictors (7):
- Age, Sex, Poverty Ratio
- Weight
- Alcohol days, Soda frequency, Sleep hours

03 TECHNICAL BACKGROUND: Support Vector Machine

Support Vector Machine(SVM) is a supervised machine learning algorithm used for classification and regression tasks. It attempts to find the best boundary known hyperplane that separates different classes in the data.

How SVM Works

For linearly seperable data, SVM identifies a hyperplane that maximizes the distance(margin) between the closest points of each class, known as support vectors. For non-linear data SVM uses kernel functions to map the data into a higher-dimensional space where separation become possible.

We optimize:
$\frac{1}{2}||w||^2 + C \sum \xi_i$
Constraint:
$y_i(w \cdot x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$

Kernels

Common kernel functions include:
- Linear kernel: which is suitable for linearly separatable data.
  $K(x_i, x_j) = x_i \cdot x_j$
- Polynomial kernel: works best in capturing interactions between features.
  $K(x_i, x_j) = (\gamma\, x_i \cdot x_j + r)^d$
- Radial Basis Function(RBF): very flexible and widely used for non- linear relationships.
  - $K(x_i, x_j) = \exp(-\gamma\, ||x_i - x_j||^2)$

Important Hyperparameters

Parameter	Description	Kernels
C (Cost)	Penalty for misclassification; higher C → tighter margin	All
γ (Gamma)	Reach of single training point; high γ → complex boundary	RBF, Poly
d (Degree)	Polynomial degree; controls boundary curvature	Polynomial

Model Evaluation

The performance any of the SVM models can be assessed using:
- **Accuracy**: tells us Out of all predictions, how many are correct? Misleading when working with imbalanced data.
- **Precision**: focuses on the event we care about identifying (i.e. Cancer in this project). A high precision indicates that predicted cancer cases are usually correct. Low precision means the model says positive too often even when it is wrong.
- **Recall**: focuses on out of the truly positive classes, how many did the model correctly catch? A low recall implies that many positives were missed while a high recall indicates very few false negatives.
- **F1-Score:** is a balance between precision and Recall.

Considerations & Limitations

Linear Kernel
- Only works well with classes that are linearly separable
- Single tuning parameter C – easiest to tune
- Interpretable
- Sensitive to class imbalance without weighing

Radial Kernel
- Most flexible – handles complex, non-linear boundaries
- Two tuning parameters (C, γ) – hence most expensive to tune.
- High γ – lead to overfitting; Low γ: leads to underfitting
- Difficult to interpret; misleadingly high accuracy under class imbalance

Polynomial Kernel
- Able to capture feature interactions using degree d
- Two tuning parameters (C, d) – could lead to overfitting.
- Less stable than RBF for similar flexibility
- Two tuning parameters (C, γ) – hence most expensive to tune.
- High γ – lead to overfitting; Low γ: leads to underfitting

All Kernels
- Feature scaling is required.
- Class weights or resampling needed for imbalanced data
- Computationally expensive for large datasets

04 Methodology

Data Source & Target Variable

The dataset comes from 2022 National Health Interview Survey(NHIS). The target variable is CANCEREV, recorded into a binary factor: 1 = Cancer, 0 = No Cancer. Invalid or unknown codes (0,7,8,9) were set to NA and those rows dropped (7,501 observations removed).

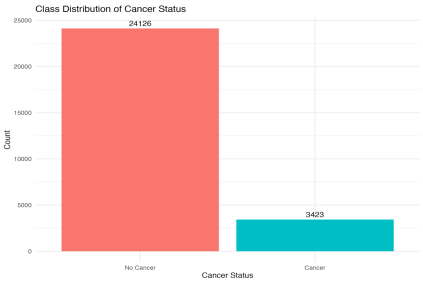
Variable Selection

- Seven predictors were selected across all three themes.
- Demographics: Age(years), Sex(male/female),and poverty ratio.
- Body Metrics: Weight(lbs)
- Lifestyle habits: Alcohol Days per year, Soda Frequency, and Sleep Hours per night.
- BMI was initially considered but removed due to collinearity with Weight.
- Smoking excluded(34K+ missing due to survey skip-pattern)

Missing Data Handling

- Age & Sex(<70 missing): rows dropped
- Weight, Alcohol, Soda, Sleep (moderate missingness): median imputation applied – robust to right skewed distribution

Class Imbalance



- ~ 88% No Cancer vs. ~12% Cancer
- Class Weights Applied: Cancer: = 7, No Cancer = 1 (weight = max count/minority count)

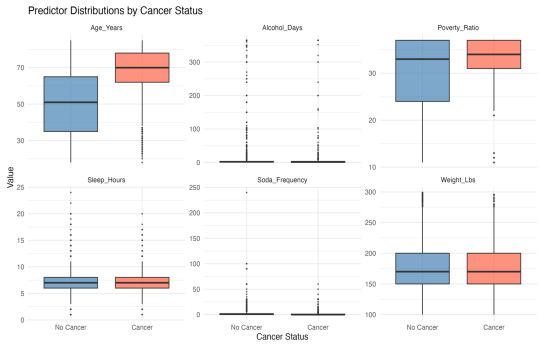
Hyperparameter Tuning

- Cross-validation on training set only.
- Cost ∈ {0.01, 0.1, 1, 10} — all kernels
- Gamma ∈ {0.01, 0.1, 1, 10} — radial kernel
- Degree ∈ {2, 3, 4} — polynomial kernel
- Best params → models refit on full training set before test evaluation

Evaluation Metrics

- Accuracy, Precision, Recall, F1-Score
- Recall = primary metric (medical screening context; missing a cancer case is the costliest error
- Accuracy noted as misleading under class imbalance

05 Results



Predictor Strength(EDA – Box Plots)

- Age : strongest predictor – median ~70(Cancer) vs. ~50(No Cancer), 20-year gap with minimal overlap
- Poverty Ratio: weak secondary signal – Cancer group shows slightly higher median and tighter spread
- Weight & Sleep hours: nearly identical distribution between groups
- Alcohol Days & Soda frequency: zero-inflated as majority report zero, making them poor predictors

Hyperparameter Tuning Results

- Linear: best C param: 0.1
- Radial: best C param: 10, Gamma: 0.1
- Polynomial: best C param: 0.1, degree = 4 (CV error = 0.310)

Model Performance on Test Set

Kernel	Accuracy	Precision	Recall	F1 Score
Linear	0.653	0.224	0.798 ✓	0.352
Polynomial ★	0.699	0.243	0.741	0.362 ★
Radial	*0.806* ↑	0.236	0.199 ✗	0.216 ✗

06 Discussion

Age is the Dominant Predictor
The SVM boundary falls between ages 55–65. The median cancer diagnosis age of ~70 vs. ~50 for non-cancer confirms epidemiological evidence that cancer risk rises sharply with age. This was consistent across all three kernels.

Poverty Ratio: Healthcare Access Signal
Both groups average 3–3.5× the poverty line. The slightly higher median in the Cancer group likely reflects better healthcare screening and diagnosis access among higher-income respondents, rather than higher true disease burden.

Poverty Ratio: Healthcare Access Signal
Alcohol days and soda frequency are zero-inflated with near-identical distributions between groups. Sleep hours and weight showed substantial overlap. These variables alone are insufficient predictors — consistent with cancer's multifactorial nature.

Poverty Ratio: Healthcare Access Signal
Radial kernel's high accuracy (0.81) is misleading — it rarely predicts Cancer, achieving only 0.20 recall. Linear kernel maximizes recall (0.80) at the cost of precision (0.22). Polynomial kernel offers the best F1 balance (0.36, recall 0.74).

Class Imbalance Challenge
With ~88% No Cancer observations, standard accuracy is an unreliable metric. Class weighting (Cancer weight = 7) helped, but further techniques like SMOTE or cost-sensitive learning could improve sensitivity.

Limitations
Missing smoking data (34k+ observations skipped survey question) prevented inclusion of a key risk factor. Survey weights not applied — results may not represent the U.S. population. SVMs lack native probability outputs.

07 Conclusion

- Age emerged as the sole dominant predictor of cancer diagnosis across all SVM kernels, with the decision boundary consistently falling between ages 55–65. Lifestyle variables — alcohol, soda, sleep, and weight — showed minimal individual discriminatory power, suggesting these factors cannot predict cancer status in isolation.
- The polynomial kernel (F1 = 0.36, Recall = 0.74) provided the best balance for this imbalanced classification task. The linear kernel (Recall = 0.80) remains preferable in a strict cancer screening context where missing a true case carries the highest cost. The radial kernel, despite its high accuracy, is not recommended without additional tuning to address class imbalance.
- For policymakers, age-targeted screening programs — particularly for individuals 60+ — represent the most evidence-supported intervention from this analysis.

- https://www.geeksforgeeks.org/machine-learning/support-vector-machine-algorithm/.
- https://www.who.int/health-topics/cancer#tab=tab_1
- https://stanfordhealthcare.org/content/shc/en/medical-conditions/cancer/cancer.html

[1] Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Backman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. https://doi.org/10.18128/D070.V7.4. Links to an external site.http://www.nhis.ipums.orgLinks to an external site..
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